DNA Methylation Analysis: Statistics and Machine Learning in Breast Cancer

Thesis Work Presentation
Master’s Degree in Mathematical Engineering

Elisabetta Roviera

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Introduction: Goals of the Presentation

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- My background: Mathematical Engineering and Data Science
- Objective: show how quantitative tools can be applied to biomedical problems
- Case study: DNA methylation in breast cancer

Breast Cancer: Context and Relevance

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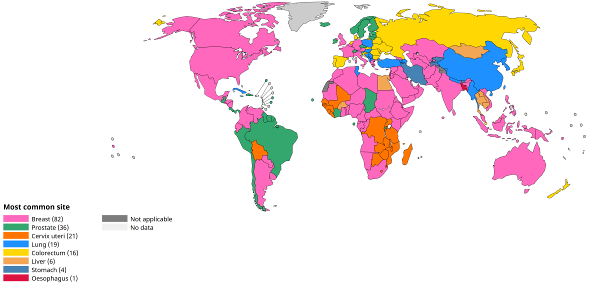
- High incidence worldwide
- Importance of early detection for patient outcomes
- Why data matters: quantitative analysis supports screening, diagnosis, and risk assessment

Most common site per country, Absolute numbers, Incidence, Both sexes, in 2022 (excl. NMSC)

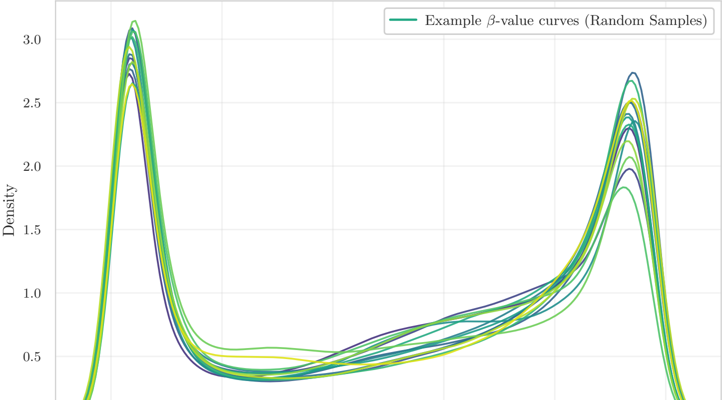


Cancer TODAY | IARC
https://gco.iarc.who.int/today
Data version: Globocan 2022 (version 1.1) - 08.02.2024

International Agency for Research on Cancer
World Health Organization

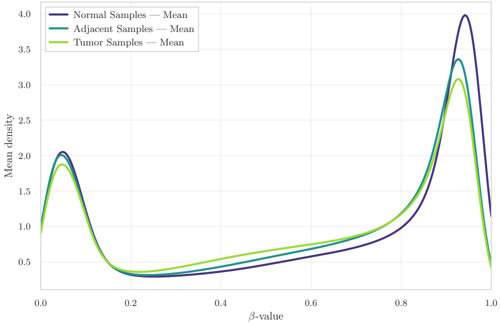
CpG and β-values: The Language of Epigenetic Data

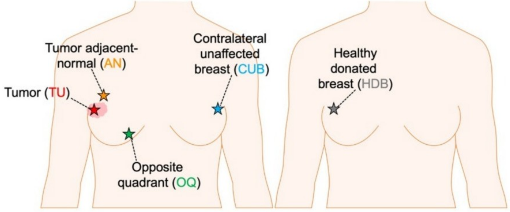
- CpG sites: genomic positions where DNA methylation can occur
- β-value $\in [0, 1]$: proportion of methylation at each CpG
- Methylation modulates gene regulation
- In cancer: both hypermethylation and hypomethylation emerge



Field Cancerization: The Central Research Question

- Is "normal" tissue near the tumor already epigenetically altered?
- As Dr. Gambino highlights:
  "Identify early methylation alterations in normal tissues and assess their role in tumor initiation."
- Concept of field cancerization





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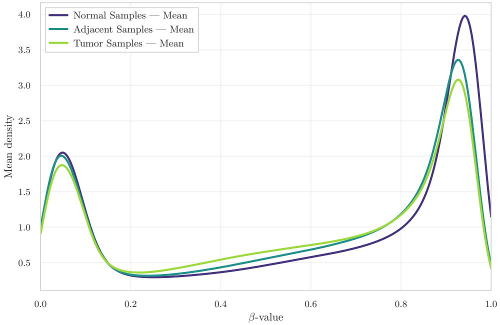

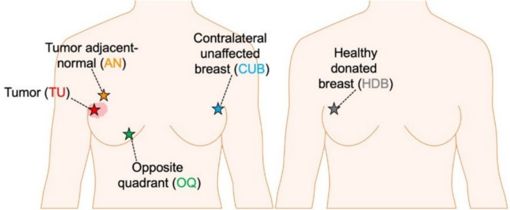

Accessing and Understanding the Data

- GEO datasets: heterogeneous formats and incomplete metadata
- Small n, huge p (up to 800k CpG sites)
- Raw IDAT files vs. preprocessed β-value matrices
- Unknown or inconsistent preprocessing in some datasets

Dataset	**Samples (n)**	**CpGs (p)**
GSE69914	397	485,512
GSE225845	477	750,426
GSE287331	311	702,573

Computational Challenges

- Huge files → use of Parquet format, LZ4 compression, chunked loading
- Strict RAM constraints on Kaggle environments
- Need for Illumina manifest alignment to correctly map probe IDs

Can We Merge the Datasets?

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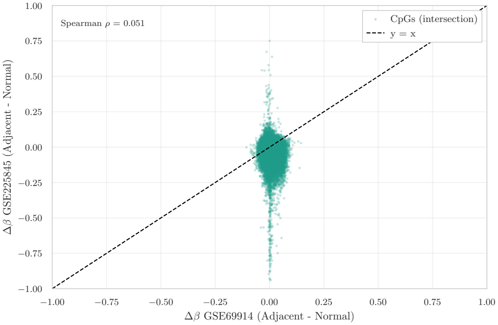
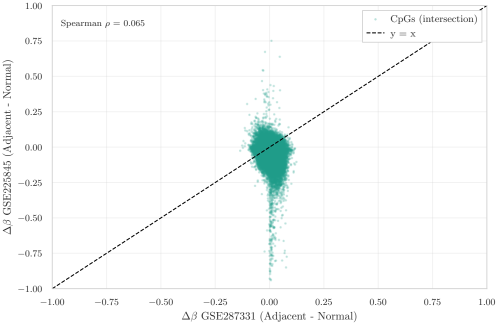
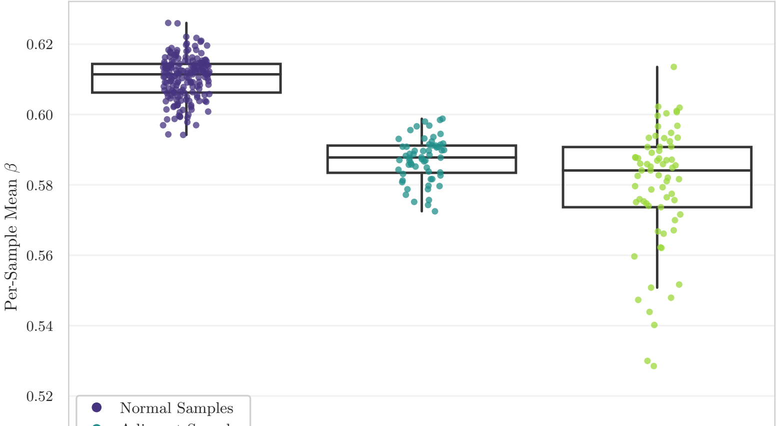
- Different array platforms (HM450 vs. EPIC)
- Heterogeneous and partly unknown preprocessing pipelines
- Direct merge would introduce strong batch and platform effects
- Decision: perform all analyses within each dataset (intra-dataset)





Global DNA Methylation Distributions

- Canonical bimodal β-value distribution (unmethylated vs. highly methylated CpGs)
- Tumor samples: shift towards hypomethylation and increased heterogeneity
- Adjacent samples: early, subtle deviations from the Normal baseline



Variability and Epigenetic Outliers

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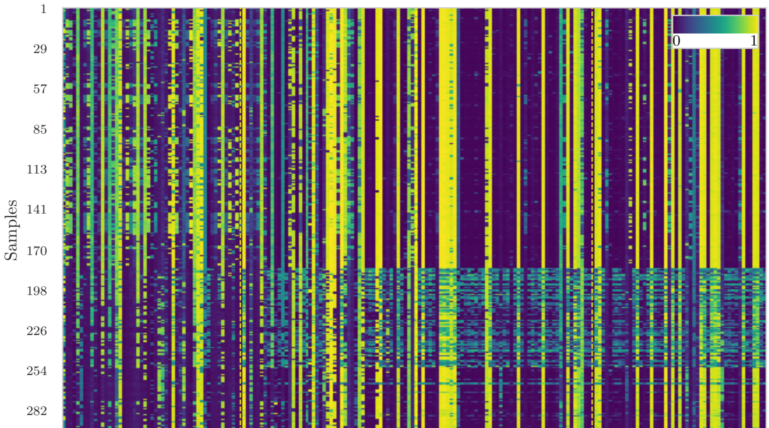
- Tumor samples show higher methylation variability
- Adjacent tissue exhibits more focal, localized signals
- Highly unstable CpGs are promising biomarker candidates



Dataset Embeddings and Group Structure

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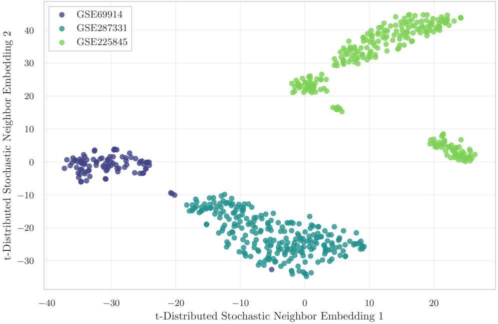
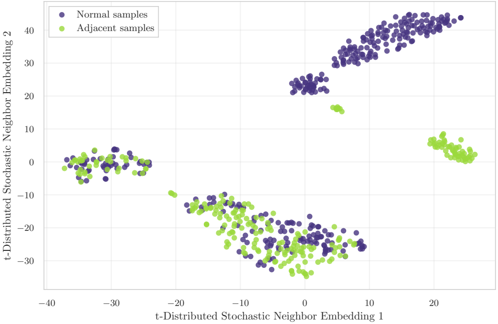
- GSE69914: strong overlap between groups
- GSE225845: more complex, diffuse patterns
- GSE287331: clear separation between Normal, Adjacent, and Tumor



t-SNE coloured by tissue state



t-SNE coloured by dataset

Probe-Design Bias and BMIQ Normalization

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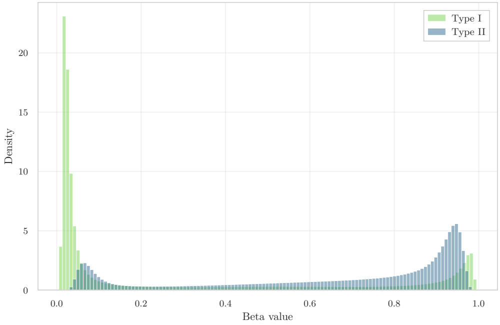
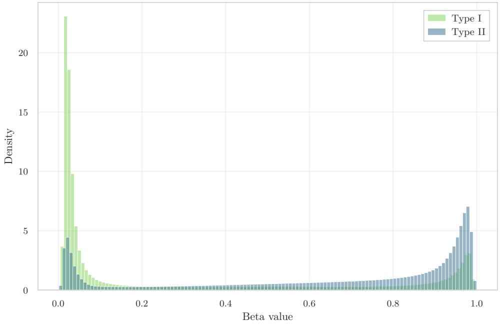
- Two probe designs (Type I vs. Type II) with incompatible β-distributions
- Real case: GSE287331 shows a strong uncorrected probe-design bias
- BMIQ performs quantile mapping of Type II probes toward the Type I distribution
- Caveat: BMIQ can reduce biological variability and partially mask real signals



Pre-BMIQ: Type I vs Type II



Post-BMIQ: Type I vs Type II

Feature Selection in High Dimensions

- $p \gg n \rightarrow$ dimensionality reduction and filtering are essential
- Technical filters: cross-reactive probes, SNP probes, non-CpG probes, Illumina manifest
- Evaluate and correct potential batch effects (e.g., COMBAT)
- Three families of feature selection strategies:
  - Statistical: $\Delta\beta$, variance filtering, stability measures
  - Biological: promoters, CpG islands, curated gene sets
  - ML-based: Lasso, decision trees, embedded model selection

700k CpGs → 300k → 150k → 3k Features →

ML in Non-Ideal Conditions

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- Train / validation / test with extremely small n
- Generalizing across datasets (A $\longrightarrow$ B) is often unreliable
- How many features to select? Which metrics to use?
- Overfitting, noise, limited interpretability
- Deep Learning: training from scratch is infeasible $\longrightarrow$ only pre-trained / transfer learning methods

Approach	Data Need	Interpretability	Robustness	Use-case
Statistical	low–medium	high	high	$\Delta\beta$, variance, QC
ML	medium–high	medium	medium	classification, FS
DL	very high	low	low	transfer learning only

Why This Matters for a Mathematician

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- Massive, high-dimensional biological datasets
- Statistical modelling under $p \gg n$
- Machine learning under real-world constraints
- Normalization, modelling assumptions, variance structure
- Full interdisciplinarity: biology, statistics, computing, modelling

Take-home Message

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- Epigenetics requires strong quantitative skills
- Mathematical and statistical techniques genuinely help biology
- Final message: research is closer than it seems

Thank you!

If you have any questions, I’d be happy to answer.